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Research Article

Host Evolution Improves Genetic Circuit Function in Complex Growth Environments

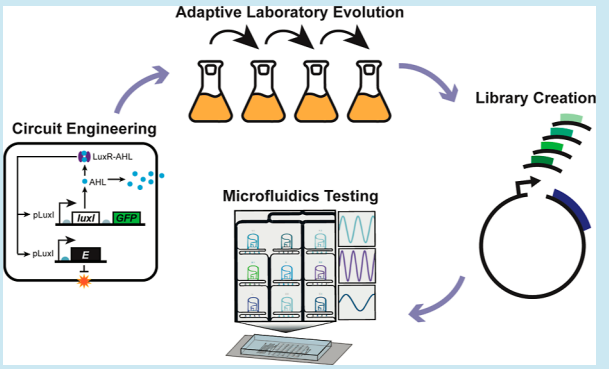
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ABSTRACT: The systematic design of genetic circuits with predictable behaviors in complex environments remains a significant challenge. Here, we engineered a population control circuit and used a combination of evolutionary and rational engineering approaches to enhance *Escherichia coli* for robust genetic circuit behavior in nontraditional growth environments. We utilized adaptive laboratory evolution (ALE) on *E. coli* MG1655 in minimal media with a sole carbon source and saw improved dynamics of the circuit after host evolution. Additionally, we applied ALE to *E. coli* Nissle, a probiotic strain, in a more complex medium environment with added reactive oxygen species (ROS) stress. In combination with directed mutagenesis and high-throughput microfluidic screening, we observed restored circuit function and improved tolerance of the circuit components. These findings serve as a framework for the optimization of relevant bacterial host strains for improved growth and gene circuit performance in complex environments.



KEYWORDS: *adaptive laboratory evolution, circuit design, strain optimization, microfluidics, microscopy*

INTRODUCTION

As genetic engineering tools improve in sophistication, synthetic biologists are able to develop increasingly complex genetic circuits for control of microbial population behavior.[1,2] These circuits have been used for the detection and treatment of cancer,[3,4] mitigation of metabolic disorders,[5] and sensing of the gut environment.[6] However, numerous limitations remain in the real-world deployment of engineered gene circuits, such as unpredictability of circuit behavior in complex growth environments.[7] Quantitative modeling of physiological transitions in *Escherichia coli* from one growth medium to another showed marked differences in metabolic activity in response to changes in medium nutrients, which can significantly influence protein production and affect underlying circuits.[8,9] Expression of synthetic gene cassettes further places stress on engineered populations, increasing sensitivity to external disturbances.[10] Thus, gene circuits that behave consistently in nutrient-rich growth conditions in the lab can oftentimes break down when transferred to more complex and nontraditional environments.[11]

Traditional tools to improve genetic circuit behavior employ rational engineering principles, where individual circuit components are chosen to meet specific requirements.[12] To combat environmental fluctuations, control engineering principles can be employed to limit the influence of external perturbations on circuit function.[13] However, these often require multiple genetic components that place a significant metabolic burden on the cells. Seminal work by Yokobayashi et al. used directed evolution of specific circuit regions to rapidly evolve a nonfunctioning circuit to regain functionality.[14] As such, the generation of mutant libraries via directed mutagenesis followed by functionality screening has become an important component of circuit development and tuning. However, this process is oftentimes time-consuming and heavily relies on trial-and-error analysis and testing.[15,16]

Adaptive laboratory evolution (ALE), where a candidate strain is cultured for a prolonged period under a selective pressure, enables the accumulation of genetic mutations for improved growth in that environment over time.[17] This technique has been used in both bacteria and yeast for the generation of desired phenotypes such as osmotic, acid, and temperature tolerance and increased alternate substrate metabolism for bioproduction.[18] Recently, ALE was used to optimize growth performance of a genome-reduced *E. coli* strain,[19] coevolve mutually auxotrophic strains of yeast and

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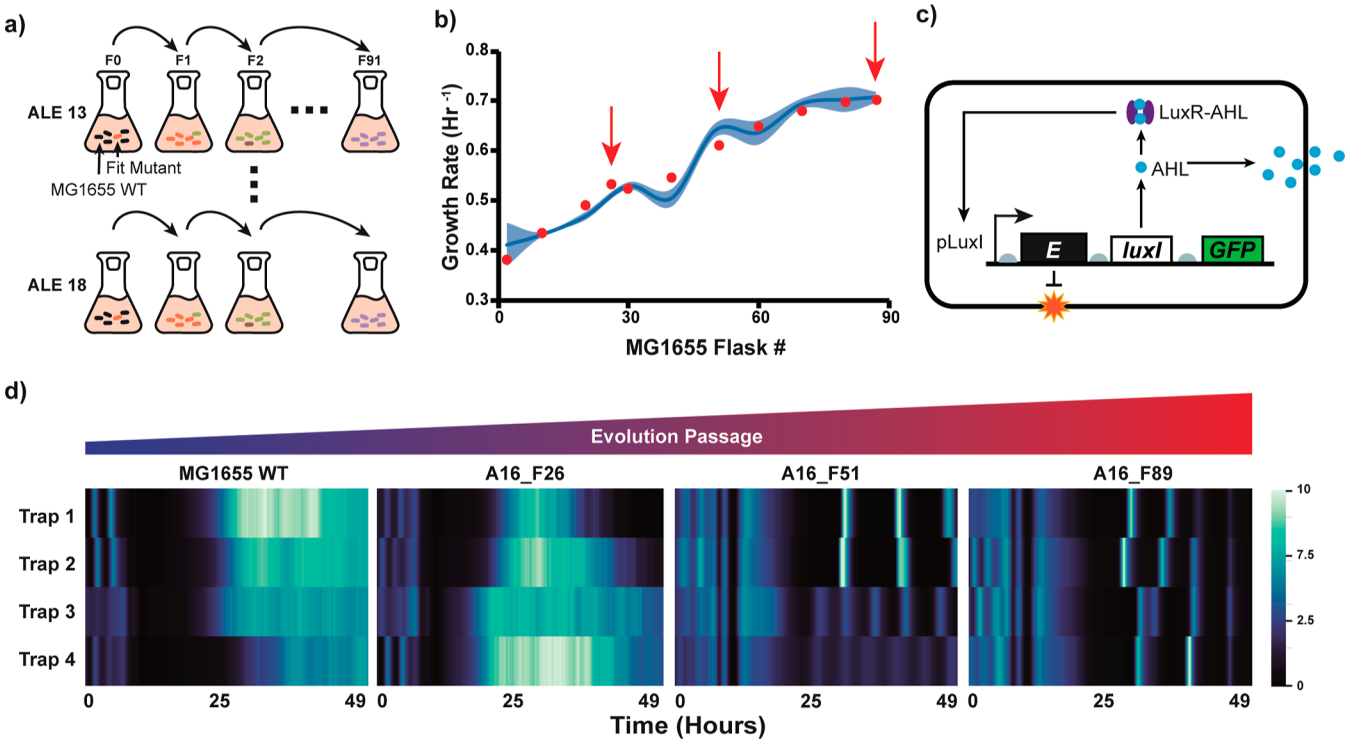


Figure 1. Evolution and characterization of MG1655 strains on M9 lactate. (a) Schematic of the ALE process. MG1655 wildtype strains were evolved in M9 lactate in 6 parallel replicate experiments (n = 6). F indicates the flask or batch number. (b) Average growth rate of the similarly performing replicates over the course of evolution. Red points indicate growth rates of A16 lineage throughout evolution. Red arrows indicate flasks that were chosen for further analysis. All data points represent mean (solid line) ± SEM (shaded areas), n = 6. (c) Diagram of the single plasmid synchronized lysis circuit (SLC). This plasmid is on high copy pUC-derived ColE1 ori. AHL production from LuxI synchronizes cells within the population and provides positive feedback. The X174E protein causes cell lysis as a negative feedback. (d) Heatmaps representing normalized GFP time traces (proxy for circuit activation) of the MG1655 wildtype and different evolved strains from one lineage with the SLC.

lactic acid bacteria for improved production of the auxotrophic compound,[20] and improve methanol utilization in *E. coli* engineered with a hybrid methanol assimilation pathway.[21] In combination with omics technologies, researchers can better understand mutational pathways underlying evolution and create strains that are tolerant to unique and stressful environmental conditions.[22] Genetic circuitry has also been incorporated with adaptive evolution, where the mutation rate is directly controlled by the concentration of a target metabolite in a feedback loop, to increase tyrosine and isoprenoid production.[23]

To date, host evolutionary engineering has been underutilized for the improvement of dynamic genetic circuits in desired environments. In this study, we leveraged the selected genetic changes from evolved strains as an alternative method for improving circuit dynamics in complex environments. As a proof of concept, we first applied selective pressure on wildtype *E. coli* K-12 MG1655 in minimal medium and found improved growth in the medium during the evolution. We chose the Synchronized Lysis Circuit (SLC), which exhibits a dynamic growth rate-dependent behavior with potential *in vivo* applications, as our candidate circuit for evaluation.[1,3,24] The SLC causes cycles of population lysis and regrowth, and we observed improvements in circuit behavior and burden tolerance after ALE. Next, we directly targeted the SLC burden tolerance of *E. coli* Nissle, a probiotic strain, through evolution of the host strain to Paraquat (PQ)-induced reactive oxygen species (ROS) stress and saw improved growth under PQ stress and recovery from circuit-induced lysis. We further increased the complexity of this evolution by using a medium base not traditionally used for bacterial culture. Using directed mutagenesis on the evolved Nissle, in combination with batch culture and high-throughput microfluidic screening, we identified a fully oscillatory strain in the complex medium, thus showing the synergy of ALE with common synthetic biology tools. Finally, we explored the differing host−circuit effects between wildtype MG1655 and Nissle using a ribosome-binding site (RBS) library. Comparative analysis revealed strain-specific responses to circuit components, thus highlighting the importance of chassis selection in synthetic biology. Overall, this work explores the complex interplay between strain choice, host evolution, circuit architecture, and medium environment in optimizing synthetic circuit performance. It provides insights into strategies for improving the robustness and predictability of synthetic circuits in non-standard conditions, paving the way for future studies on optimizing engineered biological systems for diverse applications.

RESULTS

Adaptive Laboratory Evolution in Minimal Medium Improved Genetic Circuit Behavior in MG1655. As an initial proof of concept, wildtype *E. coli* K-12 MG1655 was evolved in M9, a minimal medium, with lactate as the sole carbon source. Six replicates were evolved in parallel to examine adaptive convergence (Figure 1a).[25] In batch culture growth, we observed that the growth rate on M9 lactate increased with subsequent evolutionary passages in all

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replicates (Figure 1b). Whole genome sequencing of the evolved replicates showed that *ppsA* and *ilvH* were the most prevalently mutated across the ALE lineages (Table S1). Both *ppsA* (phosphoenolpyruvate synthase in the gluconeogenesis pathway, found in 3 out of 5 replicates) and *ilvH* (acetolactate synthase subunit, found in 2 out of 5 replicates) were found to be mutated in previous ALE experiments in M9 lactate.[26−28] As the ALE16 lineage showed representative growth rates throughout the evolution, clones from this replicate at multiple stages of the evolution were chosen for further analysis of the circuit behavior (Figure 1b).

To study the effect of the evolved host strain on the circuit response, we chose to characterize the synchronized lysis circuit (SLC), as it requires the combination of multiple circuit components and is sensitive to environmental factors and growth rate. In the SLC, expression of the LuxI protein leads to the production of quorum sensing autoinducer *N*-Acyl homoserine lactone (AHL) for synchronous population behavior. Upon reaching a threshold population size and AHL concentration, the *pLux* promoter drives expression of the lysis protein E from phage X174 (X174E), leading to synchronized population lysis. Following lysis, a few cells survive and reseed the population, leading to continuous cycles of growth and lysis. The circuit has been used for the continuous production and delivery of cancer therapeutics within tumors.[3,24,29] Thus, improving its behavior is an important area of study. Wildtype MG1655, A16-F26 (ALE16-Flask26), A16-F51, and A16-F89 strains were transformed with a single plasmid version of SLC (Figure 1c). In order to eliminate interexperimental variations, we chose to use a multistrain microfluidic device that can accommodate up to 48 unique strains of *E. coli* for testing (Figure S1). Strains in each trap were grown in LB for 11 h to establish baseline circuit dynamics, followed by a medium change to M9 lactate. As M9 lactate completely replaced LB, both wildtype MG1655 and A16-F26 exhibited one last lysis peak at 7 h after medium change, followed by a prolonged delay before regrowth without distinct lysis dynamics. Meanwhile, both A16-51 and A16-89 regained consistent lysis cycles following the medium change (Figure 1d, Video S1). Sequencing analysis revealed that F51 and F89 clones shared mutations in *ppsA* as well as in *hfq*, an RNA chaperone protein that binds to mRNA for translational regulation under environmental and metabolic stress (Table S1). Mutations in *hfq* were present in previous ALE experiments in M9 lactate and may contribute to reducing cellular stress response for adaptation.[26] These results suggest that mutations mainly involve metabolic changes in response to M9 lactate and should not confer any constitutive growth or circuit behavior advantages in a nonselective LB environment. Indeed, we observed no significant differences in circuit behavior between the wildtype and the most evolved A16-F89 strain in the initial LB medium (Figure S2a).

We hypothesized that the improvement in circuit dynamics of A16-F51 and A16-F89 was due to improved growth and increased burden tolerance of the host strain in M9 lactate from the evolution. After the final population lysis at 7 h, the wildtype and F26 strains showed delayed regrowth in M9 lactate, most likely due to impaired postlysis recovery and a reduced growth rate in the nutrient-limited medium. Previous work has shown that the X174E protein facilitates lysis through inhibition of cell wall synthesis and requires active cell growth for lysis that may be stalled in M9.[30−32] Thus, while the cells may reach quorum and activate *pLux* as seen by strong GFP expression, the process of cell lysis mediated by X174E might be metabolically expensive and requires specific cellular growth states that are compromised under nutrient limitation in M9 lactate. Meanwhile, sequencing and batch culture growth rate results suggest that the accumulated genetic mutations of F51 and F89 strains allowed for enhanced growth and metabolic activity in the M9 lactate environment. This led to improved recovery from lysis, allowing continuous cycles of growth and lysis, as seen in microfluidics (Figure 1d, Video S1). Further, we studied promoter sensitivity in an M9 lactate batch culture by testing each strain with a plasmid with the *pLux* promoter driving GFP expression. Comparing GFP expression relative to OD during the exponential phase (OD 0.4 to 0.6), we saw that A16-F89 showed improved inducibility to 1 nM AHL compared to the wildtype (Figure S2b). Interestingly, we also noticed that upon AHL induction, the wildtype strain exhibited a greater reduction in growth compared to A16-F89 (Figure S2c). Since GFP expression also places burden on the cells, it appears that the evolved strain could better handle this burden in M9 lactate. Although further research is required to fully elucidate the role of each mutated gene on circuit behavior in M9 lactate, our results demonstrate the utility of ALE for improving dynamic circuit response in stressful and nutrient-limited growth environments.

Evolution in Paraquat Improved *E. coli* Nissle Tolerance to Lysis Circuit Burden. Although *E. coli* K-12 MG1655 is widely used for development and testing of new synthetic circuits due to its favorable growth rate and easy transformability, it becomes a less suitable host candidate when considering therapeutic applications. Among various bacterial strains explored, *E. coli* Nissle, a probiotic currently in use for treatment of gastrointestinal disorders, has shown promise in clinical applications due to its favorable safety profile and relevance in areas such as inflammatory bowel disease (IBD), toxic gut metabolite accumulation, and treatment and detection of cancers.[33−36] As such, we sought to use ALE to improve circuit behavior in *E. coli* Nissle in a more complex and nontraditional growth environment. We chose DMEM, a medium formulation used for culturing mammalian cells, for our base medium, with additional modifications to include a pH of 6.5, 0.4 mM glucose, and 30 mM lactate. As we noticed that ALE improved circuit burden tolerance in the proof-of-concept study in MG1655 (previous section), we hypothesized that Nissle could be directly adapted for the faster recovery of survivors following SLC lysis. Recent work shows that the X174E protein prevents peptidoglycan synthesis,[30] which can lead to production of reactive oxygen species (ROS) stress.[37] Thus, the evolution of Nissle to tolerate higher levels of ROS may further impact its SLC behavior. As such, we chose to also include 350 μM of Paraquat (PQ), a redox cycling compound that is able to generate high levels of superoxide ROS by removing electrons from cellular electron carriers and reducing oxygen, in the DMEM.[38,39]

Five Nissle ALE replicates were evolved in parallel to generate an optimized host strain for evaluation of the SLC. We observed an increase in the growth rate within the first 30 passages, after which the rate began to plateau (Figure S3a). Each replicate lineage was sequenced for mutations at the end of the evolution in order to capture the adaptive mutational response. We found that *argP* and *rpoA* were the most prevalent mutations across the ALE lineages. Mutations in *argP*, which controls transcription of genes related to arginine and lysine metabolism, were found in 2 out of 5 replicates,

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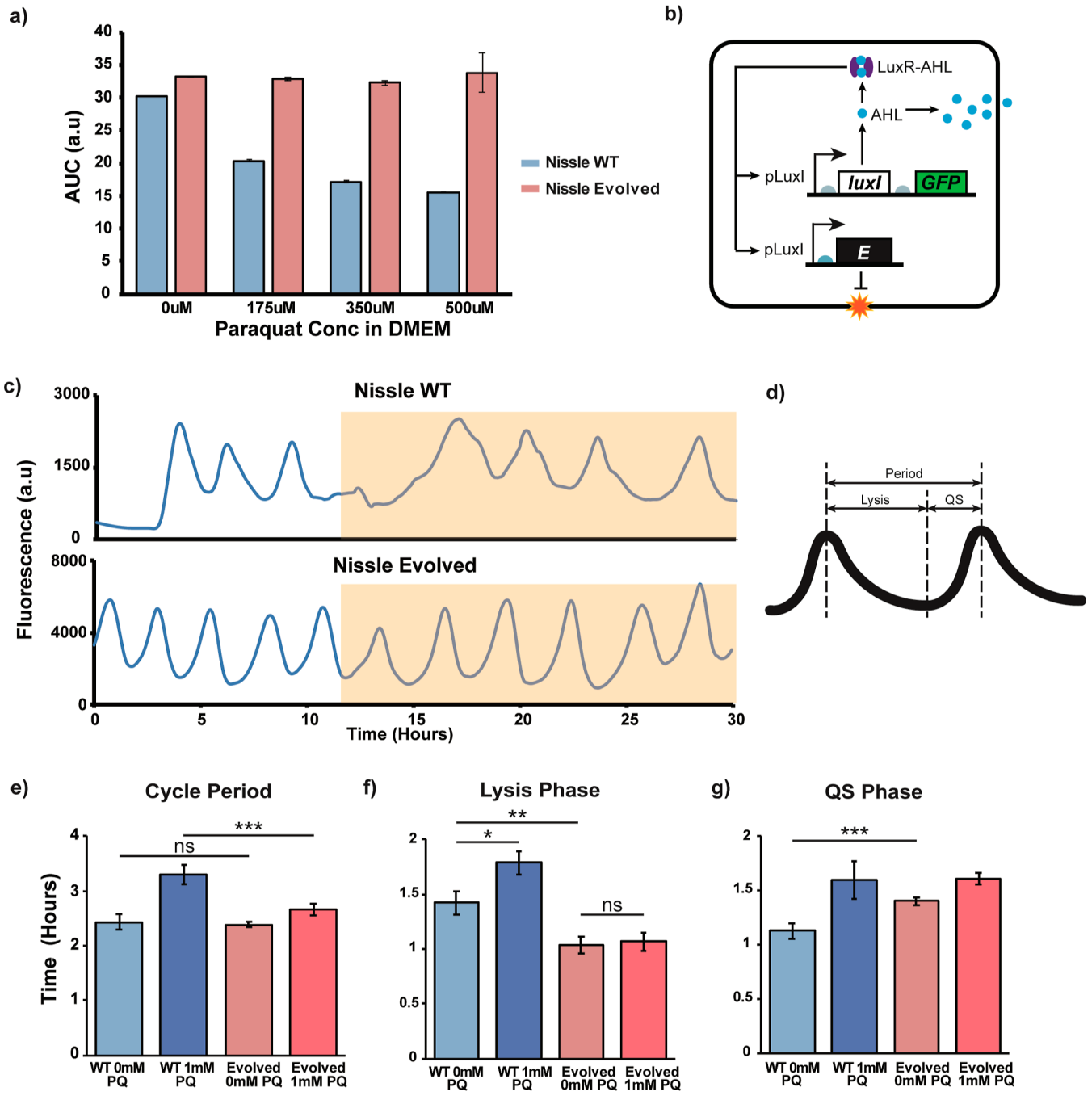


Figure 2. Characterization of the wildtype and evolved Nissle strain response to ROS stress. (a) Comparison of the wildtype and evolved Nissle growth in the DMEM-based evolution medium in the presence of increasing Paraquat concentrations. (b) Diagram of two-plasmid SLC. In this version, LuxI and GFP are on a high copy pUC-derived ColE1 ori plasmid, while X174E is on a medium copy p15A ori plasmid. (c) Representative time traces of GFP fluorescence of the wildtype and evolved Nissle with SLC growing in microfluidic traps in a continuous culture of LB. The yellow shaded region indicates introduction of 1 mM Paraquat. (d) Schematic of SLC characterization. Each cycle of GFP oscillation is broken down into Lysis and Quorum Sensing (QS) phases for comparison. (e–g) Comparison of the cycle period, lysis phase period, and QS phase period for the wildtype and evolved Nissle before and after 1 mM Paraquat introduction in microfluidics. (e) $P_{ns} = 0.62$ ($N_{WT} = 14$, $N_{evolved} = 57$), $^{***}P = 0.0015$ ($N_{WT} = 16$, $N_{evolved} = 40$). (f) $^{*}P = 0.0213$ ($N_{0mM} = 21$, $N_{1\ mM} = 19$), $^{**}P = 0.0062$ ($N_{WT} = 21$, $N_{evolved} = 60$), $P_{ns} = 0.7807$ ($N_{0\ mM} = 60$, $N_{1\ mM} = 44$). (g) $^{***}P = 0.0008$ ($N_{WT} = 15$, $N_{evolved} = 60$).

suggesting evolution to the growth medium base. Meanwhile, mutations in *rpoA*, a transcriptional subunit that was found to affect anaerobically expressed genes in *Salmonella*,[40] were found in 3 out of 5 replicates (Table S1). These genes are different from a previous study utilizing ALE and multiomics analysis for the impact of PQ stress on *E. coli* MG1655 physiology, most likely due to a lower PQ concentration, a significantly different evolution growth medium, and a different strain subtype.[41] As the ALE2 lineage displayed a representative growth phenotype, a clone from Flask114 was chosen for further analysis (Figure S3a). We will hereafter refer to the ALE2-Flask114 strain as the evolved Nissle strain.

The evolved Nissle host displayed a differential improvement compared to the wildtype strain in the evolution growth

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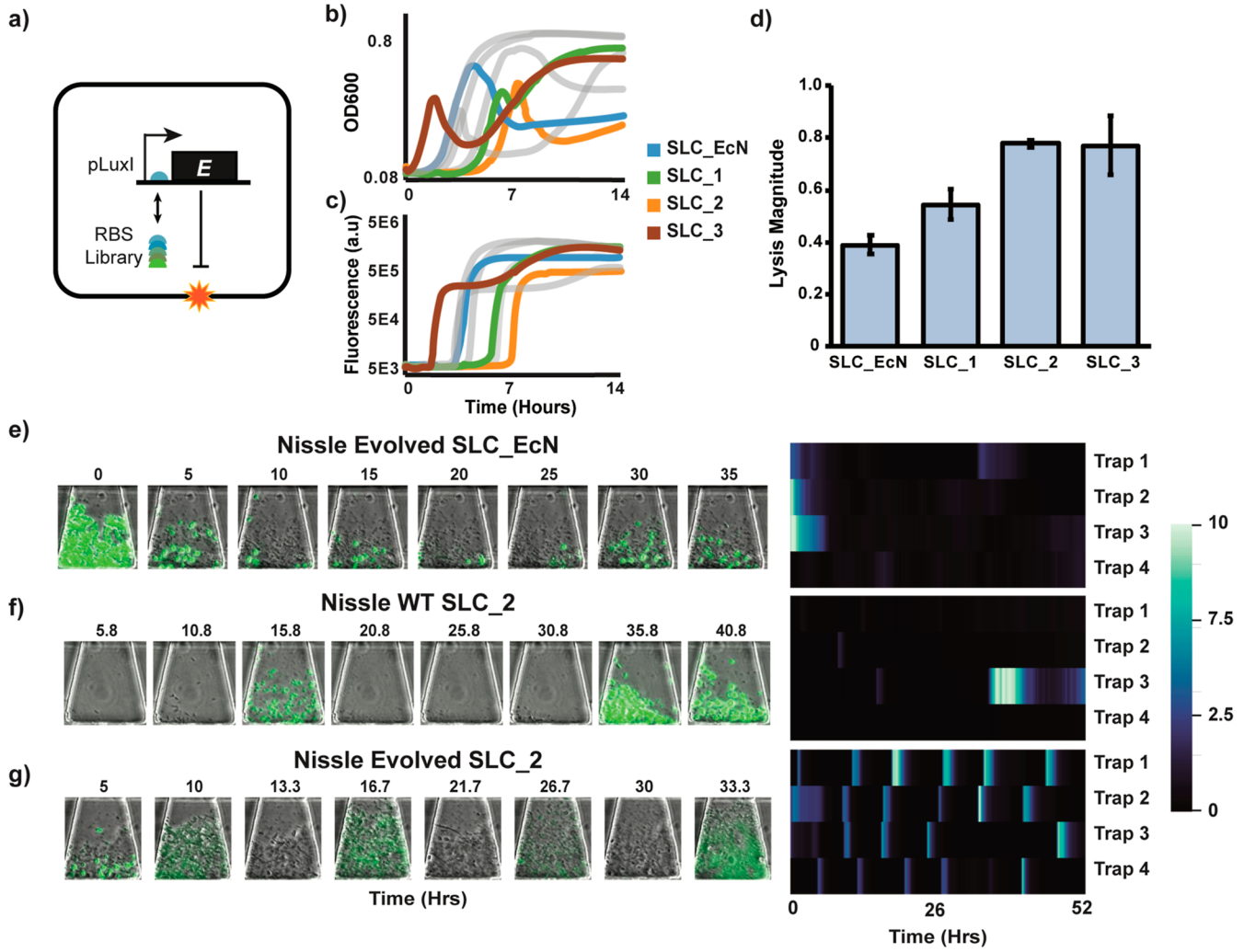


Figure 3. Optimization and characterization of wildtype and evolved Nissle circuit dynamics in a complex DMEM-based medium. (a) Schematic of SLC library generation via directed mutagenesis of the X174E protein ribosome-binding site (RBS). (b,c) Batch culture results of the SLC library. Both OD_{600} and GFP fluorescence were characterized over 14 h. Three SLC versions were chosen that represented the range of SLC behaviors. (d) Comparison of lysis magnitudes associated with each RBS. Bars represent mean ± SEM (n = 3). (e−g) Representative time series images from transmitted light and fluorescence channels of evolved Nissle with original SLC-EcN, wildtype Nissle with optimized SLC-2, and evolved Nissle with optimized SLC-2 in a complex DMEM-based medium. Corresponding heatmaps of GFP time series data for each strain.

medium environment. We detected only a small improvement in growth in the evolved strain compared to the wildtype strain in the DMEM-based medium without PQ. However, when increasing levels of PQ stress were introduced, a proportional reduction in growth for wildtype Nissle was observed, whereas evolved Nissle was unchanged from the nonstress condition (Figures 2a, S3b). As we noticed nonlog-linear growth in the batch culture which prevented accurate measurement of a single growth rate, likely due to multiple substrate utilization phases from the complex DMEM, we chose to quantify the area under the curve (AUC) of the growth curves. These findings show that the evolved strain did not exhibit a trade-off in the growth rate in the unstressed conditions while conferring ROS stress resistance.

For our candidate circuit, we chose to characterize a two-plasmid version of the SLC that was previously used in *Salmonella* for delivery of cancer therapy as our candidate circuit.[1] In this circuit, *pLux* driving X174E is on a medium copy p15A origin plasmid and *pLux* driving LuxI and GFP is on a high copy pUC-derived ColE1 origin plasmid (Figure 2b). Previous work integrated the SLC into the Nissle genome for therapy delivery; however, its dynamics were not characterized in detail in microfluidics.[3] We developed a two-plasmid version (SLC-EcN) that functions in Nissle (Table S2). Since the evolution conditions included both an alternate medium and PQ-induced ROS stress, we chose to characterize the circuit behavior in response to each condition individually.

We first studied the effect of the evolution to PQ on SLC dynamics. In order to isolate circuit response to only PQ, we chose to first characterize dynamics in LB. Both the wildtype and evolved strains were transformed with the SLC-EcN circuit. Both strains were then tested in the same multistrain microfluidic device. The strains were first grown in LB for 11 h to obtain at least 3 lysis peaks for establishing baseline circuit dynamics, followed by the addition of 1 mM PQ for a controlled comparison of circuit behavior dynamics before and after the introduction of PQ-induced stress (Figure 2c, Video S2). We chose to use 1 mM PQ in order to place both the wildtype and evolved strains under significant ROS stress during the microfluidic analysis, as preliminary batch culture

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growth assays indicated the evolved strain's growth was largely unaffected by 500 μM PQ (Figure 2a). We saw similar oscillation cycle periods in both strains growing in LB without PQ (Figure 2e). We took a closer look at the population dynamics by breaking each cycle into a lysis phase and quorum sensing (QS) phase for comparison. The lysis phase consists of the period from the highest GFP fluorescence to the lowest fluorescence due to population death, while the QS phase consists of the period from the lowest GFP fluorescence to the highest fluorescence due to AHL accumulation after lysis recovery (Figure 2d).

In LB without PQ, the lysis phase of the wildtype strain was significantly longer than that of the evolved strain, suggesting the wildtype strain was more affected by X174E and required longer recovery before restarting growth (Figure 2f). Testing in the batch culture, where *pLux* only drives X174E, showed that after lysis induction with 10 nM AHL, the wildtype strain exhibited less robust recovery compared to the evolved strain (Figure S4a,b). Further, when testing with *pLux* driving GFP, we saw similar GFP induction magnitudes between the two strains, suggesting both strains showed promoter sensitivity similar to that of AHL (Figure S4c). Thus, evolution to PQ-induced ROS was able to improve recovery of lysis survivors in Nissle. Meanwhile, in the QS phase, when grown without PQ, we noticed that the wildtype strain required significantly less time to reach quorum compared to the evolved strain (Figure 2g). Previous work shows that *E. coli*'s native oxidative stress control system can interact with the Lux quorum sensing system.[42−44] While further testing is needed to elucidate the exact mechanisms by which PQ-induced ROS stress evolution affects different quorum sensing components, our results demonstrate that host evolution can impact the circuit response in favorable growth conditions.

After 11 h of growth, 1 mM PQ was added to the media. After PQ addition, the cycle period increased for both strains, but the wildtype strain oscillated at a significantly longer period compared to the evolved strain, suggesting that although both strains were affected by the ROS induced by PQ, the effect was reduced in the evolved strain (Figure 2e). We saw an increase in the QS phase for both strains but less so for the evolved strain (Figure 2g). Meanwhile, the lysis phase significantly increased in the wildtype strain, whereas the evolved strain remained unaffected, further confirming that PQ evolution improved Nissle recovery from the lysis component of the circuit (Figure 2f). Since we observed both genomic differences between wildtype and evolved strains as well as growth rate differences in PQ in the batch culture without any circuit addition, we believe adaptation and response to PQ is intrinsic to the cell. Thus, any changes in the circuit behavior after PQ addition is due to PQ affecting cellular processes and should not be affected by the time point at which PQ is introduced to the culture.

Our results demonstrate that evolution to PQ led to a significant improvement in circuit dynamics when exposed to ROS. Further, the mutations accumulated from this evolution influenced Nissle response to different SLC circuit components that produce or are influenced by ROS. Notably, we observed that improved recovery from circuit-induced lysis in the evolved Nissle strain persisted when cultured back in a standard LB medium, as evidenced by a significantly shorter lysis phase compared to the wildtype (Figure 2f). This indicates that adaptation to PQ-induced ROS stress conferred a constitutive change in the circuit behavior in nonselective conditions, potentially due to the direct role of ROS stress in the lysis circuit. The increased tolerance to X174E and subsequent reduction in the lysis recovery period could be important for therapy delivery in cancer applications.

ALE in Combination with RBS Library Screening Improved Circuit Behavior in the Evolution Growth Medium. Although we only observed a small improvement in the Nissle growth rate in the complex DMEM-based medium without PQ after evolution, the evolved strain contained a mutation related to the metabolism of lysine and arginine (*argP*), both of which were found in the medium (Table S1). We hypothesized that we would still see differences in SLC behavior between the wildtype and evolved strain in the DMEM-based medium without PQ. As expected, the wildtype strain with SLC did not oscillate in the DMEM-based medium (Figure S5a). However, the evolved strain with SLC was also unable to oscillate, exhibiting only one large lysis peak, after which the cells were unable to recover, with only a few cells remaining in a fluorescent state (Figure 3e, Video S3). We attributed this to significant changes in growth medium conditions, which may alter the metabolic needs of the cells and their response to different components of the circuit.

We hypothesized that SLC dynamics in the DMEM-based medium can be optimized by tuning the production of X174E (Figure 3a). Using a ribosome-binding site (RBS) generator as well as sequences found in previous work, we generated a library of RBSs of X174E using directed mutagenesis[1,45] (Table S2). In LB, growth in the batch culture showed distinct dynamics in growth and GFP production (Figure 3b,c). Three SLC versions, SLC-1, SLC-2, and SLC-3, reflected different batch culture dynamics and were chosen for further analysis. To examine these RBSs and their potential effects on the SLC, we isolated the "lysis" part of the circuit by removing AHL production, in which each RBS is placed in front of X174E which is driven by *pLux*. Lysis magnitude was calculated for each variant at 10 nM of AHL induction (Figure S5b). SLC-EcN yielded the smallest lysis magnitude. SLC-1 showed the second strongest lysis, followed by SLC-2 and SLC-3 with similar lysis magnitudes (Figure 3d).

We found that when the evolved Nissle strain was transformed with SLC-2, we were able to again see consistent lysis dynamics (Figures 3g, S5c). The cycle period was significantly longer in the DMEM-based medium, with a period of 8.93 $\pm$ 0.72 h compared to 2.38 $\pm$ 0.05 h in LB (Figure S5c). Meanwhile, the wildtype strain with SLC-2 exhibited only 1 or 2 lysis peaks after which it was unable to recover in growth. Further, the wildtype strain had trouble growing in the microfluidic traps, most likely due to burden imposed by the circuit (Figure 3f). We confirmed this in the batch culture in the DMEM-based medium, again using *pLux* driving X174E. After 10 nM AHL induction of lysis, growth recovery was less robust in the wildtype strain after normalizing by growth without lysis (Figure S6a,b). Further, the wildtype strain harboring SLC-2 took a significantly longer time to begin growth compared to the evolved strain (Figure S6c). Finally, when we studied the response of each strain to AHL using *pLux* driving GFP, we saw that at 10 nM AHL induction, the evolved strain produced more GFP compared to the wildtype (Figure S6d).

Thus, although we observed only a small difference in the growth rate between the wildtype and evolved strains in the batch culture (Figure 2a), the accumulated mutations from ALE improved growth and behavior with the SLC (Figure 3g).

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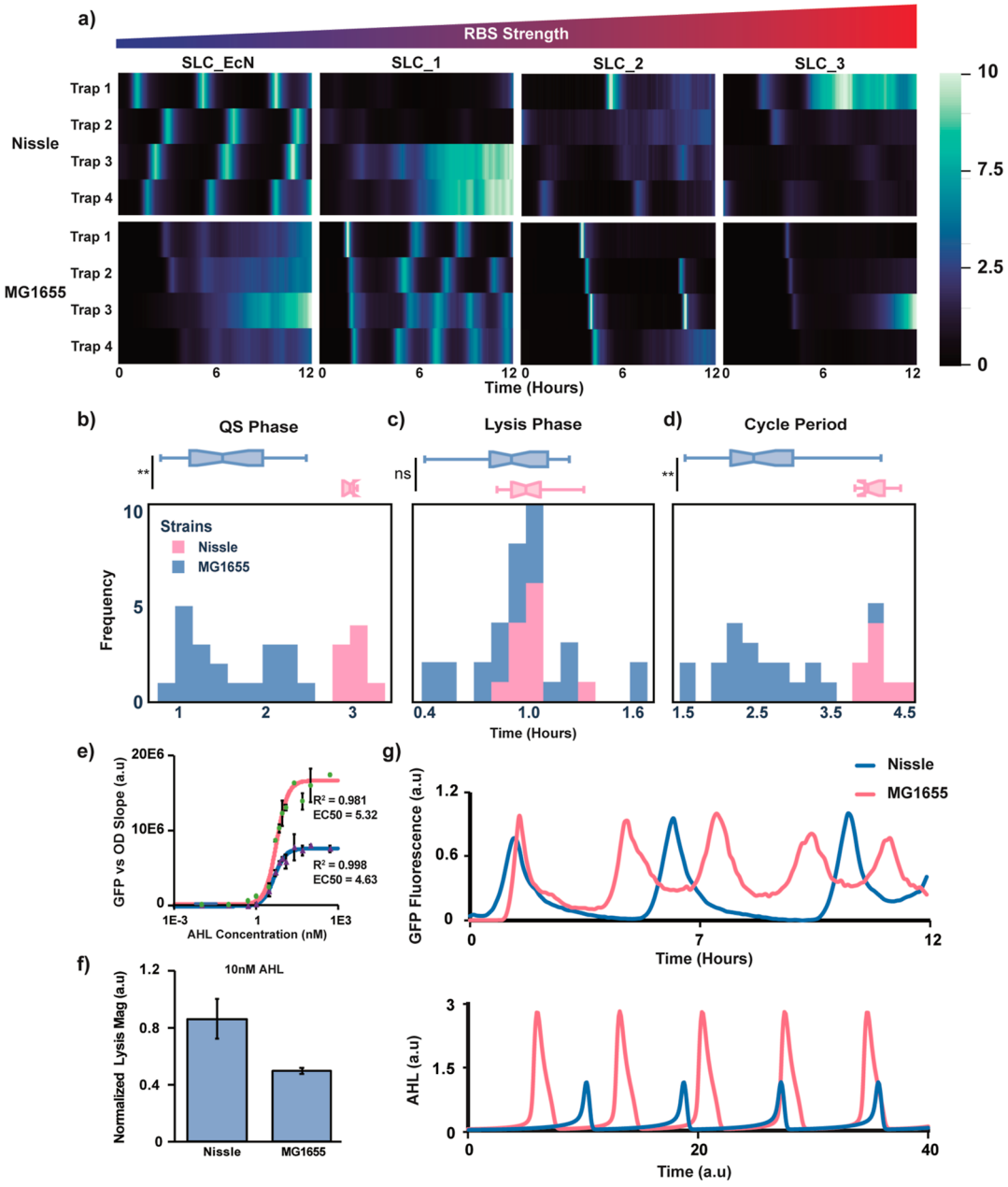


Figure 4. Characterization of wildtype Nissle and MG1655 SLC population dynamics. (a) Heatmaps of GFP time series data of wildtype Nissle and MG1655 with different SLC RBS library versions in microfluidics. (b–d) Comparison of the QS phase, lysis phase, and cycle period of Nissle and MG1655. Top: Boxplots of each period distribution. Bottom: Histogram of each period distribution. (b) $^{**}P = 0.0001$ ($N_{\text{Nissle}} = 8$, $N_{\text{MG1655}} = 20$). (c) $P_{\text{ns}} = 0.5319$ ($N_{\text{Nissle}} = 12$, $N_{\text{MG1655}} = 24$). (d) $^{**}P = 0.0001$ ($N_{\text{Nissle}} = 8$, $N_{\text{MG1655}} = 20$). (e) Dose–response curve of AHL vs GFP induction response. Data points represent mean ± SEM ($n = 3$). The solid line represents the fitted Hill function. (f) Comparison of lysis susceptibility between MG1655 and Nissle. Bars = mean ± SEM ($n = 3$). (g) Top: representative time series GFP data of MG1655 and Nissle SLC dynamics. Bottom: Computer simulations of the mathematical model for each strain.

Further, our results show that although ALE ensures improved growth and adaptation to complex growth environments, circuit parameters may also need to be retuned for robust circuit dynamics.

Comparison of Nissle and MG1655 Circuit Behavior. During our investigation, we noticed different circuit responses between Nissle and MG1655 due to inherent genomic differences between the two strains. Previous comparative studies between Nissle and other *E. coli* strains showed that Nissle shared only 73.2% of homologous genes with MG1655.[46] Although the growth behavior of Nissle has been well studied through metabolic network reconstruction, little work has been done to compare synthetic circuit response between Nissle and MG1655.[47] Therefore, we sought to gain further insights into the differences in the circuit behavior between wildtype MG1655 and Nissle by focusing on SLC dynamics. We chose to use the wildtype strains for this comparison in order to eliminate any differences that may be caused by genetic mutations acquired during ALE. Since we did not see consistent oscillations of wildtype Nissle in the DMEM-based medium, we performed our comparisons in LB.

We used the SLC variants developed in the previous section for circuit dynamic comparisons. Wildtype Nissle strains harboring each SLC variant were examined in further detail

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on a microfluidic chip (Table S2). Compared to consistent oscillations as seen with SLC-EcN, an increase in RBS strength in SLC-1 led to no growth or growth without oscillations. Further increase in RBS strength (SLC-2) led to more variability between traps, with one or two lysis events before loss of oscillations. SLC-3 yielded dynamics similar to those of SLC-2, with the population generally exhibiting one strong lysis event followed by weak or complete loss of oscillatory behavior after regrowth (Figure 4a).

We next transformed the same SLC variants into wildtype MG1655. As expected, MG1655 exhibited a significantly different behavior to each SLC variant compared to Nissle. Whereas SLC-EcN yielded consistent oscillations in Nissle, we saw extremely weak or no oscillations in MG1655 followed by continuous growth in the traps. Meanwhile, an increase in RBS strength in SLC-1 led to consistent oscillations in all traps. All traps of SLC-2 showed one strong lysis event, with two traps showing two lysis events before loss of oscillations. In SLC-3, one strong lysis event, followed by a loss of oscillatory behavior, was observed. Overall, we observed that MG1655 was able to oscillate at stronger RBS strengths (Figure 4a).

We next compared the dynamics between Nissle with SLC-EcN and MG1655 with SLC-1 by quantifying each part of the SLC cycle. We found that the QS phase in MG1655 was significantly shorter compared to that in Nissle, contributing to a significantly faster lysis cycle (Figure 4b,d). We examined the responsiveness of the *pLux* promoter of each strain using a gradient of AHL to induce *pLux* driving GFP expression. We quantified the slope of the OD vs GFP plots at log growth following induction to get an estimate of GFP production per cell. After fitting to a Hill function, we saw that both strains had similar EC50s, suggesting that *pLux* in both strains has similar sensitivity. However, we saw that the GFP expression level of MG1655 was consistently higher (2-fold higher response at 10 nM AHL induction), suggesting that for the SLC, MG1655 has a higher LuxI production rate than Nissle, leading to a faster accumulation of AHL and a shorter time to reach the quorum threshold (Figure 4b,e). When we compared the Lysis Phase, we saw no significant difference between the two strains (Figure 4c). We next examined each strain's susceptibility to the X174E protein using the *pLux* promoter driving X174E instead of GFP. After induction with 10 nM AHL, we saw the lysis magnitude of Nissle was higher than that of MG1655 after normalizing by protein production (Figure 4f). We derived a simple deterministic model of SLC dynamics (Supporting Information). Using this model, we replicated the difference in SLC dynamics observed between the two strains (Figure 4g). Further analysis shows that as lysis sensitivity increases, the period and QS phase increase, while the Lysis phase decreases (Figure S7a). Meanwhile, all three metrics decrease as g, which is inversely proportional to AHL production, increases (Figure S7b). These results highlight how inherent genomic differences play an important role in host−circuit response, and chassis selection must be taken into account during circuit design.

DISCUSSION

Despite continuous development of tools for rational engineering of synthetic circuits,[48−50] the need for trial-and-error testing and mass screening remains a challenge in the creation of complex circuits.[7,16] This problem is further exacerbated in the move toward real-world deployment, where circuits developed in the lab must now function predictably in poorly characterized and new growth environments. This study demonstrated the feasibility of combining ALE and traditional circuit engineering approaches for the transition from lab-originated design to translational application, where an intricate circuit can be made to regain function in a complex growth environment. In an M9 minimal medium with lactate, MG1655 was able to regain SLC dynamics through the host evolution alone. Meanwhile, evolution in a complex DMEM-based medium not only significantly reduced the effect of Paraquat-induced ROS stress on Nissle but also improved its tolerance of burdensome SLC components. In combination with directed mutagenesis of the SLC, the evolved Nissle with SLC was able to fully oscillate in the complex DMEM-based medium environment. Notably, we found that evolution to M9 lactate did not confer any constitutive changes to the circuit behavior in nonselective LB media, while evolution to PQ did lead to a constitutive improvement in circuit-induced lysis recovery. This contrast suggests that the type of evolutionary pressure applied (adaptation to a specific growth source versus a general stressor such as PQ) may impact the generalizability of the resulting mutations on the circuit performance in different environments. For example, adaptation to PQ might confer broader resilience against ROS-generating circuits or environments, whereas adaptation to a specific carbon source likely yields benefits primarily when that nutrient is available. While a number of previous studies have mainly focused on using adaptive laboratory evolution for improving growth rates of bacteria and yeast for bioproduction, this work leverages ALE's power for robust synthetic circuit function. Further, as both *E. coli* Nissle- and quorum sensing-based systems are becoming increasingly popular in clinical applications, this work provides a detailed comparison and modeling of SLC dynamics between MG1655 and Nissle.

The methods presented in this study can be applied to the development of a wide variety of circuits in a range of host strains, given the generalizability of evolutionary engineering techniques. One immediate example is in the realm of bacterial cancer therapy, where strains such as *E. coli* and *Salmonella typhimurium* can potentially be anaerobically evolved for improved growth, circuit response, and therapy delivery in the hypoxic tumor core.[51,52] While the genotypes of the evolved optimized clones were determined, further work is necessary to understand the impact of specific mutations on the overall phenotype. Such studies are informative but often require significant reverse engineering and omics analyses to elucidate mutational mechanisms.[53] Furthermore, we note that interpretation of lysis cessation of wildtype MG1655 in the M9 medium (Figure 1d) was based on reduced growth and circuit burden tolerance compared to evolved strains. Direct comparison of plasmid integrity and lysis protein function between wildtype and evolved strains following prolonged growth in microfluidics remains an area for future investigations. Finally, although we were able to compare SLC dynamics between wildtype Nissle and MG1655 in LB, we were unable to adapt both strains to the same environment due to resource limitations. Such studies may reveal inherent differences in adaptability between different strains as well as subsequent changes in the circuit behavior, which further underscores the importance of host strain selection for performance in specific environments. Overall, this study provides a framework for the incorporation of evolutionary engineering and rational synthetic circuit design for the

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development and eventual deployment of complex genetic circuits.

MATERIALS AND METHODS

Adaptive Laboratory Evolution. ALE was performed in 15 mL of evolution media at 37 °C and well mixed at 1100 rpm for full aeration. An automated system passed the culture to fresh flasks before they had reached the stationary phase (Tecan Sunrise plate reader, equivalent to OD_{600} = 1 on a traditional spectrophotometer with 1 cm path length). A maximum specific growth rate (h^{-1}) was calculated by linear regression of ln(OD_{600}) and time (h) during the exponential growth phase (OD_{600} = 1). For *E. coli* MG1655, the evolution media consisted of M9 minimal media with 2 g/L lactate (Sigma, CAT 60389). Six replicates were evolved in parallel for the identification of common mutation targets. For *E. coli* Nissle, the evolution media consisted of DMEM (Sigma, CAT D5030), 30 mM lactate (Sigma, CAT 60389), 0.4 mM glucose, 70 mM MES (Sigma, CAT 69889), 1 g/L NH_4Cl, and 350 μM Paraquat (Thermo, CAT AC227320050), with pH adjusted to 6.4 using NaOH or HCl as needed. Five replicates were evolved in parallel.

DNA Sequencing and Mutation Calling. DNA from each replicate at the end point of the evolution lineage was isolated from an overnight culture and centrifuged for 5 min at 8000 rpm. The cell pellet was frozen at −80 °C. Genomic DNA was isolated using the Quick-DNA Fungal/Bacterial Microprep Kit (Zymo Research) following manufacturer's protocol, including RNase A treatment. Sequencing libraries were prepared using a Kapa HyperPlus Kit (Roche Diagnostics) following manufacturer's protocol. Libraries were run on HiSeq or NextSeq (Illumina). Sequencing reads were filtered and trimmed using AfterQC version 0.9.7.75. We mapped reads to the *E. coli* K-12 MG1655 reference genome (NC_00913.3) and *E. coli* Nissle reference genome (NZ_CP022686) and cryptic plasmids (NZ_CP058219, NZ_MW240712) primarily using breseq version 0.33.1.5 as a part of ALEdb's mutation calling platform.[54] Mutation analysis was performed using ALEdb Web site.[55]

Plasmids and Strains. MG1655 and Nissle strains (wildtype and evolved) were cultured in either lysogeny broth (LB) media or evolution media. Appropriate antibiotics were added for the plasmid maintenance. Plasmids pSpSLC_RBS11, pTD103-sfGFP, and pZA-X174E-pTacHlyE were constructed by our group in previous studies.[1,16] pZA-X174E-EcN was obtained by replacing the ribosome-binding site of X174E with that of pZA-X174E-pTacHlyE (Table S2). Plasmids were transformed into all MG1655 strains via chemical transformation and Nissle strains via electroporation (Table S3).

Library Creation for Evolved Nissle SLC Strains. *E. coli* Nissle A2-F114 was first made electrocompetent, transformed with pTD103-sfGFP via electroporation, and plated on an LB agar plate containing kanamycin. The next day, one colony of Nissle containing pTD103-sfGFP was selected, grown up overnight, and made electrocompetent.

To generate a mutant library of the pZA-X174E-EcN plasmid, a ribosome-binding site library calculator was used for site-directed mutagenesis primer design.[45] The entire plasmid was PCR amplified with the following degenerate primers where *N* indicates any base: 5′ TAATAGATATAAATANG-GANNTTGNC ATGGTACGCTGGACTTTG 3′ and 5′ ATTCGACTATAACAAACCATTTTC 3′. The PCR was incubated with DPNI at 37 °C for 1 h to digest the template plasmid. The resulting PCR product (3.1 kb) was run on an agarose gel and extracted using a QIAquick Gel Extraction Kit (QIAGEN). Blunt-end DNA ligation was performed by mixing 4 μL of the gel-extracted PCR product with 0.5 μL T4 ligase buffer, 0.5 μL T4 PNK, and 1 μL of DNase-free water and incubated at 37 °C for 1 h. Next, 0.5 μL of T4 ligase buffer, 0.5 μL of T4 DNA ligase, and 4 μL of DNase-free water were added to the reaction mixture and incubated at room temperature overnight. The next day, 50 μL of chemically competent *E. coli* DH5alpha cells was transformed with 5 μL of the reaction mix and plated on an LB agar plate containing 0.2% glucose and chloramphenicol. The entire plate was scraped into 5 mL of LB to harvest all colonies, and pooled library plasmids were isolated using the QIAprep Spin Miniprep Kit. Following sequencing confirmation of library creation, the pooled library plasmids were electroporated into electrocompetent *E. coli* Nissle strains containing pTD103-sfGFP and plated on an LB agar plate containing 1% glucose, kanamycin, and chloramphenicol for 15 h. 48 colonies from the agar plate were randomly selected for mutant screening and grown up for 16 h in LB media with 1% glucose, kanamycin, and chloramphenicol prior to use in experiments. 2 μL of each strain was added to 200 μL of fresh media with antibiotics in a standard Falcon tissue culture 96-well flat bottom plate. Cells were incubated at 37 °C with shaking in a Tecan Infinite M200 Pro. The OD at 600 nm absorbance and GFP fluorescence (488 nm excitation, emission 520 nm) was measured every 5 min.

Multistrain Microfluidic Device Fabrication and the Experimental Protocol. A polydimethylsiloxane (PDMS) device was made from the microfluidic device wafer by mixing 77 g of Sylgard 184 and pouring it on the wafer centered on a level 5″ × 5″ glass plate surrounded with an aluminum foil seal. The degassed wafer and PDMS were cured on a flat surface for 1 h at 95 °C. We used a multistrain microfluidic device that was previously developed in the lab.[16] The device consists of a 6 × 8 array of cell-trapping regions that can be loaded with liquid bacterial cultures. Each position contains four smaller cell traps downstream of the large trapping region that serve as regions of interest (ROIs) for tracking population dynamics in fluorescent and transmitted light channels. Up to 48 distinct positions can be loaded with a unique *E. coli* strain for continuous culture and data acquisition at high spatiotemporal resolutions.

Cells were grown overnight on LB media with appropriate antibiotics and 1% glucose to suppress expression of the *pLux* promoter driving lysis.[56] 200 μL of each strain was spun down and resuspended in 50 μL of media with 0.075% Tween-20. A PDMS device and 4″ × 3″ glass slide were cleaned with 70% Ethanol and exposed to oxygen plasma. Each strain was transferred to a cell-trapping region via handspotting using a 30 gauge syringe tip. The device and glass slide were bonded together and cured at 37 °C for 2 h. Following curing, the device was placed in a vacuum for 30 min and then mounted onto a custom optical enclosure. The inlet port was connected to a 30 mL syringe and Tygon tubing with 15 mL of LB media, antibiotics, and 0.075% Tween-20 (used as a surfactant to reduce clogging). The waste port was connected to a 30 mL syringe and Tygon tubing with 1 mL of Milli-Q water. The height difference between the inlet and outlet was 2″ corresponding to a flow rate of approximately 0.5 mL/h. After traps were filled to confluence, the LB medium was

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removed in entirety and replaced with 15 mL of evolution media containing appropriate antibiotics and 0.075% Tween-20.

Imaging and Data Extraction. Images were acquired with a Nikon TI2 instrument using a Photometrics CoolSnap cooled charge-coupled device (CCD) camera. The microscope was housed in a plexiglass incubation chamber maintained at 37 °C by a heating unit. Phase-contrast images were taken at 10× magnification at 100 μs exposure time. Fluorescent exposure times were 200 μs at 30% intensity for GFP. Images were taken every 5 min for every experiment. Fluorescence intensity profiles were extracted by analyzing frames from the GFP fluorescent channel. Images were processed in ImageJ, where a custom ROI manager was used to extract mean fluorescence values within each trap. Fluorescence values were normalized by subtracting the local background signal.

SLC Cycle Analysis. After data extraction, functions within the SciPy signal processing toolbox were used. Fluorescence traces were first smoothed using the savgol_filter. Peaks and troughs were then identified using the argrelextrema function in the SciPy signal processing toolbox, filtering for peaks and troughs with a minimum prominence of 8 (arbitrary fluorescence units). Within each cycle, the cycle period was calculated as the peak-to-peak time, the growth phase was calculated as a previous trough subtracted by the current peak time, and the lysis phase was calculated as the current peak subtracted by following the trough time.

SLC Library Transformation into Wildtype Nissle and MG1655 Strains. SLCs of interest were identified in the evolved Nissle strains via the batch culture and microfluidic analysis. The entire pZA-X174E-Lib plasmid in each SLC strain of interest was PCR amplified using the following primers: 5′ AATCGCCAGCGGCATCA 3′ and 5′ CAGGTTCATCATGCCGTCTGT 3′. The PCR reaction was processed and religated as described in the "Library Creation for Evolved Nissle SLC Strains" section. The religated PCR was transformed into DH5alpha and plated on the LB agar with chloramphenicol and 0.2% glucose. The next day, one colony was selected and grown overnight for plasmid extraction. Following sequencing confirmation, the pZA-X174E-Lib plasmid was cotransformed with pTD103-sfGFP into wildtype Nissle or MG1655.

Batch Culture Experiments. For comparison of the wildtype and evolved strain response to evolution media, the appropriate strains were seeded from a −80 °C glycerol stock into 3 mL of the evolution media and appropriate antibiotics and incubated in a 37 °C shaking incubator. The following day, 2 μL of overnight culture was added to 200 μL of fresh evolution media with appropriate antibiotics and Paraquat in a standard Falcon tissue culture 96-well flat bottom plate. Cells were incubated at 37 °C shaking in a Tecan Infinite M200 Pro. The OD at 600 nm absorbance and GFP fluorescence (488 nm excitation, emission 520 nm) was measured every 5 min. For Nissle growth rate calculation, the area under the curve (AUC) of the OD_{600} curve was calculated for the first 10 h of growth.

GFP Induction Calculation. A plasmid with *pLux* promoter driving GFP expression (M4) was transformed into strains of interest. 2 μL of each strain was added to 200 μL of fresh evolution media or LB with spectinomycin and the desired AHL concentration. Each well was monitored in a Tecan plate reader for 12 h for both OD_{600} and GFP. OD_{600} vs GFP was plotted for each well. For calculating GFP expression of MG1655 in M9 lactate (Figure S2), the AUC of OD vs GFP curves was measured. The ratio of AUCs between 1 nM and 0 nM AHL induction was calculated for each strain to get the relative AUC for comparison. For calculating GFP expression of Nissle, the OD vs GFP between OD_{600} of 0.4 and 0.55 was fitted to a linear regression and the slope was calculated. For characterizing textitpLux dose−response between MG1655 and Nissle (Figure 4), AHL concentrations between 0.01 nM and 500 μM were added and the growth response was monitored as described. A dose response curve was generated with the OD−GFP slope vs AHL concentration by fitting the data to the Hill Equation.

Lysis Magnitude Calculation. To experimentally determine relative expression strength of RBS sequences from the SLC library (Figure 3d), pZA-X174E-Lib plasmids were isolated from each SLC strain of interest as described in the "SLC Library Transformation into Wildtype Nissle and MG1655 Strains" section. These plasmids consist of pLux promoter driving X174E, where lysis is inducible by AHL. After confirmation by sequencing, each plasmid was transformed into *E. coli* DH5alpha and plated on the LB agar with 0.2% glucose and chloramphenicol. One clone was isolated and grown up overnight. 2 μL of each overnight culture was added to 200 μL of fresh LB with chloramphenicol in triplicates in a 96-well flat bottom plate and monitored for OD_{600} in a Tecan microplate reader at 37 °C with orbital shaking. Once cultures reached OD 0.3, the well plate was quickly removed from the microplate reader and AHL was added to achieve a final concentration of 10 nM. The well plate was then reinserted into the Tecan and cultures were grown for 12 h. The growth curve for each well was examined for the first inflection point where the derivative of the culture OD with respect to time changed from positive to negative. The lysis magnitude was calculated as L/G, where G is the positive change in OD from the initial time point to the inflection point, and L is the negative change in OD from the inflection point to the time point at which OD was lowest following the inflection point. To compare susceptibility to the X174E between MG1655 and Nissle (Figure 4), the lysis magnitude of Nissle at 10 nM AHL was normalized by multiplying with $\frac{\text{GFP Induction}_{\text{MG1655}}}{\text{GFP Induction}_{\text{Nissle}}}$, where GFP induction is calculated in the "GFP Induction Calculation" section at 10 nM AHL induction.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssynbio.5c00168.

Details on microfluidic chip design, sequencing data, additional strain characterization data, and simulation results and videos of microfluidic chip runs (PDF)

Table S1: All filtered mutations identified in ALE strains used in this paper (XLSX)

Video S1: Microfluidic dynamics of *E. coli* MG1655 wildtype, A16-F26, A16-F51, and A16-F89 strains with single plasmid SLC growing in M9 lactate medium (MP4)

Video S2: Microfluidic dynamics of *E. coli* Nissle wildtype and evolved strains with SLC-EcN growing in LB medium before and after 1mM Paraquat addition (MP4)

Video S3: Microfluidic dynamics of *E. coli* Nissle evolved strain with SLC-EcN and SLC-2, and *E. coli* Nissle

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wildtype strain with SLC-2, all growing in DMEM-based medium (MP4)

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Author Contributions

[⊥]A.M.F. and J.H. denote cosenior authorship. JTZ, AL, AMF, and JH contributed to project conceptualization. Adaptive laboratory evolution and sequencing were performed by MW, ECO, and AMF. All empirical experiments were performed by JTZ, AL, PE, and AS. Plasmid design and strain creation were performed by JTZ and AL. Computational modeling was performed by JTZ. This article was drafted by JTZ with contributions of revision by AL, AMF, and JH. All the authors read and approved the final manuscript.

Notes

The authors declare the following competing financial interest(s): JH declares that he is a co-founder of GenCirq Inc, which focus on cancer therapeutics. He is on the Board of Directors and has equity in GenCirq. His spouse is employed part time by GenCirq for bookkeeping and employee support with Human Resources.

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Supporting Information:

Host evolution improves genetic circuit function in complex growth environments

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Supplementary Figures and Tables

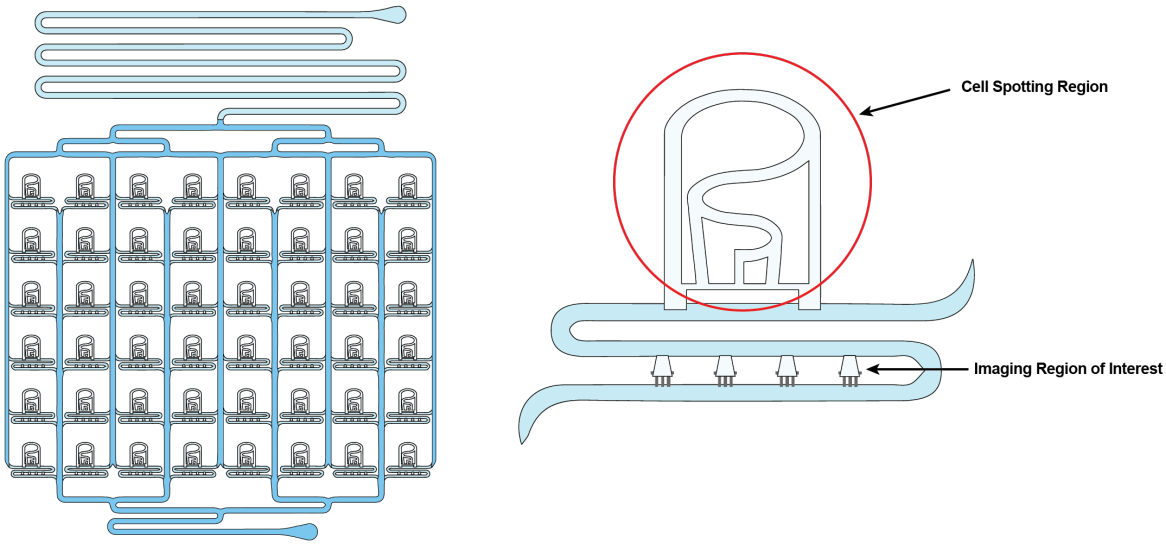


Figure S1. Device layout of a previously developed multi-strain microfluidic chip[1] The device consists of 6 by 8 array of cell-trapping regions that can be loaded with liquid bacterial cultures via hand-spotting. Four smaller cell traps downstream serve as regions of interest for tracking.

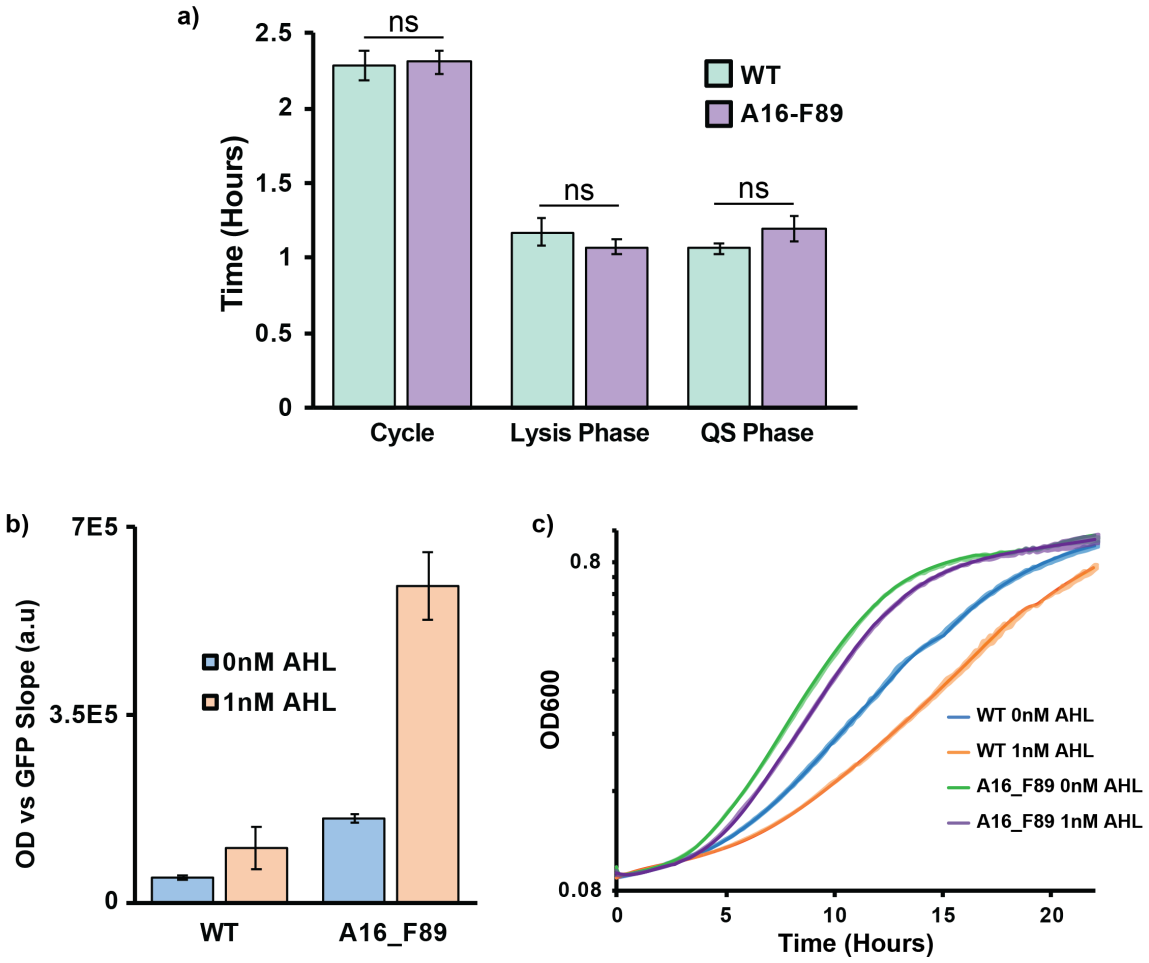


Figure S2. Characterization of MG1655 wildtype and A16-F89 strain SLC circuit behavior in LB and response to GFP induction using 10nM AHL. **a)** Comparison of SLC cycle dynamics. Cycles were analyzed according to Fig 2d. $P_{Cycle} = 0.85$ ($N_{WT} = 17$, $N_{F89} = 16$), $P_{Lysis} = 0.32$ ($N_{WT} = 18$, $N_{F89} = 18$), $P_{QS} = 0.14$ ($N_{WT} = 17$, $N_{F89} = 16$). **b)** Comparison of GFP induction at 1nM AHL addition. Bars represent mean ± SEM (n = 3). **c)** Comparison of OD600 growth curves with 1nM AHL addition. All data points represent mean (solid line) ± SEM (shaded areas), n = 3.

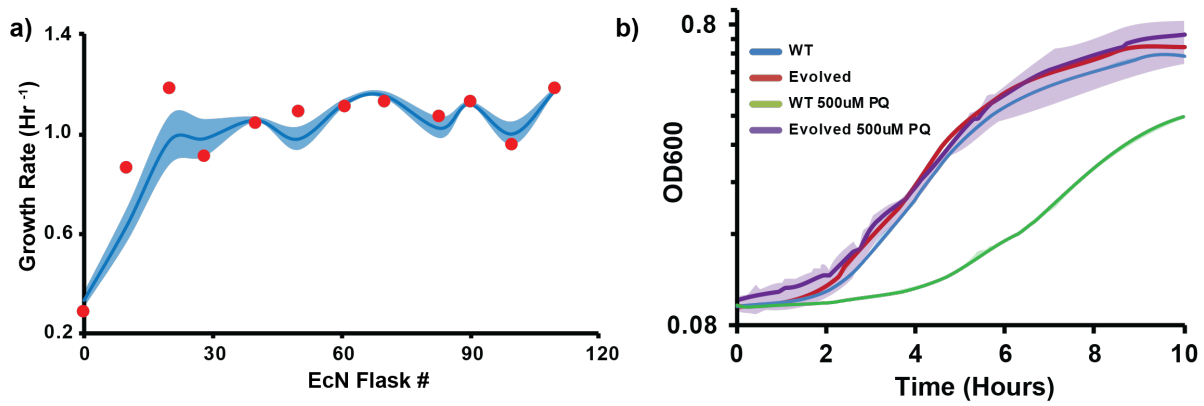


Figure S3. Characterization of Nissle growth during ALE. **a)** Growth rate of Nissle over course of ALE in complex media. Red points indicate growth rates of A2 lineage throughout evolution. All data points represent mean (solid line) ± SEM (shaded areas), n = 5. **b)** Growth curve comparison of wildtype and evolved Nissle in LB with and without 500μM Paraquat. All data points represent mean (solid line) ± SEM (shaded areas), n = 3.

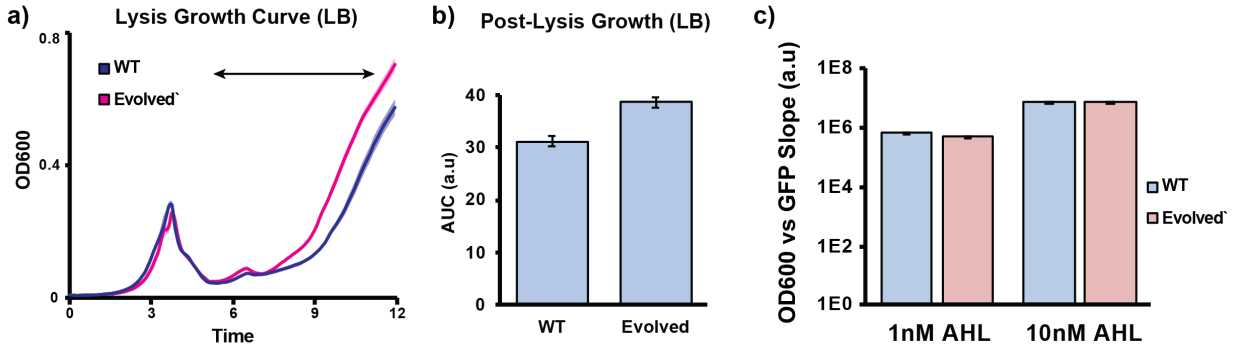


Figure S4. Batch culture characterization of lysis recovery and quorum sensing in wildtype and evolved Nissle in LB. **a)** OD curves after lysis induction for lysis recovery quantification. The AUC from the lowest OD point to six hours after that point was calculated for post-lysis growth. **b)** Comparison of post-lysis growth in LB. c) Comparison of GFP production after AHL induction. Bars represent mean ± SEM (n = 3).

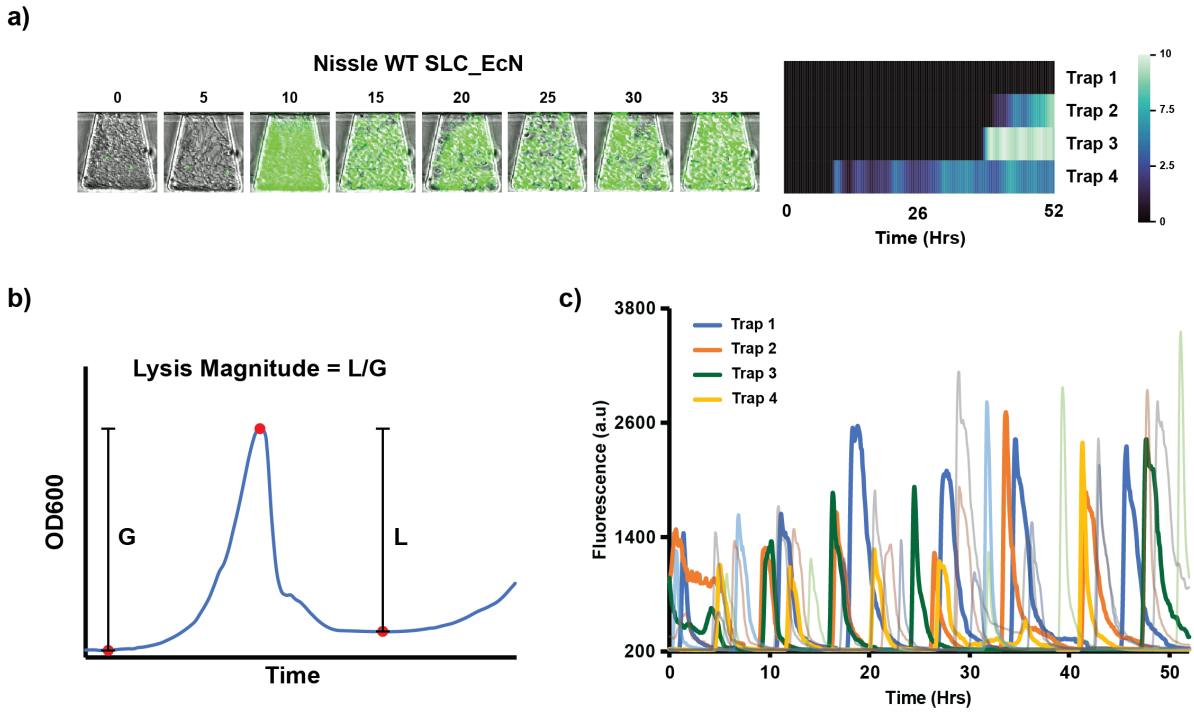


Figure S5. a) Representative time series images from transmitted light an fluorescence channels of wildtype Nissle with SLC-EcN in complex DMEM-based medium. Corresponding heatmap of normalized GFP time serires data. **b)** Diagram of lysis magnitude calculation for different RBS strengths. **c)** GFP fluorescence time traces of all individual traps of Nissle Evolved SLC-2 showing N=40 cycles of lysis. Time traces corresponding to heatmap data in Fig 3g are highlighted.

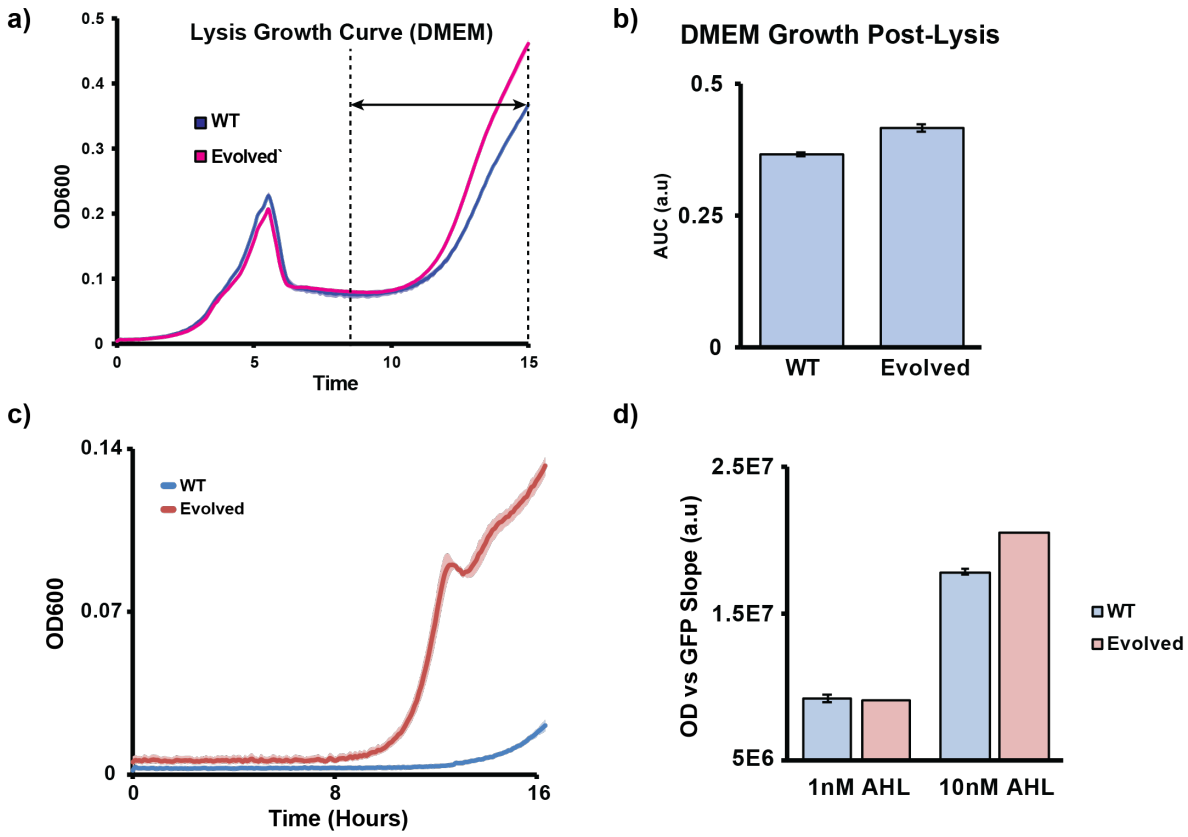


Figure S6. Batch culture characterization of wildtype and evolved Nissle in DMEM-based evolution media. **a)** OD curves after lysis induction for lysis recovery quantification. The AUC from the lowest OD point to six hours after that point was calculated for post-lysis growth. **b)** Comparison of post-lysis growth. Bars represent mean ± SEM (n = 3). **c)** Comparison of batch culture growth with SLC-2. All data points represent mean (solid line) ± SEM (shaded areas), n = 3. **d)** Comparison of GFP production after AHL induction. Bars represent mean ± SEM (n = 3).

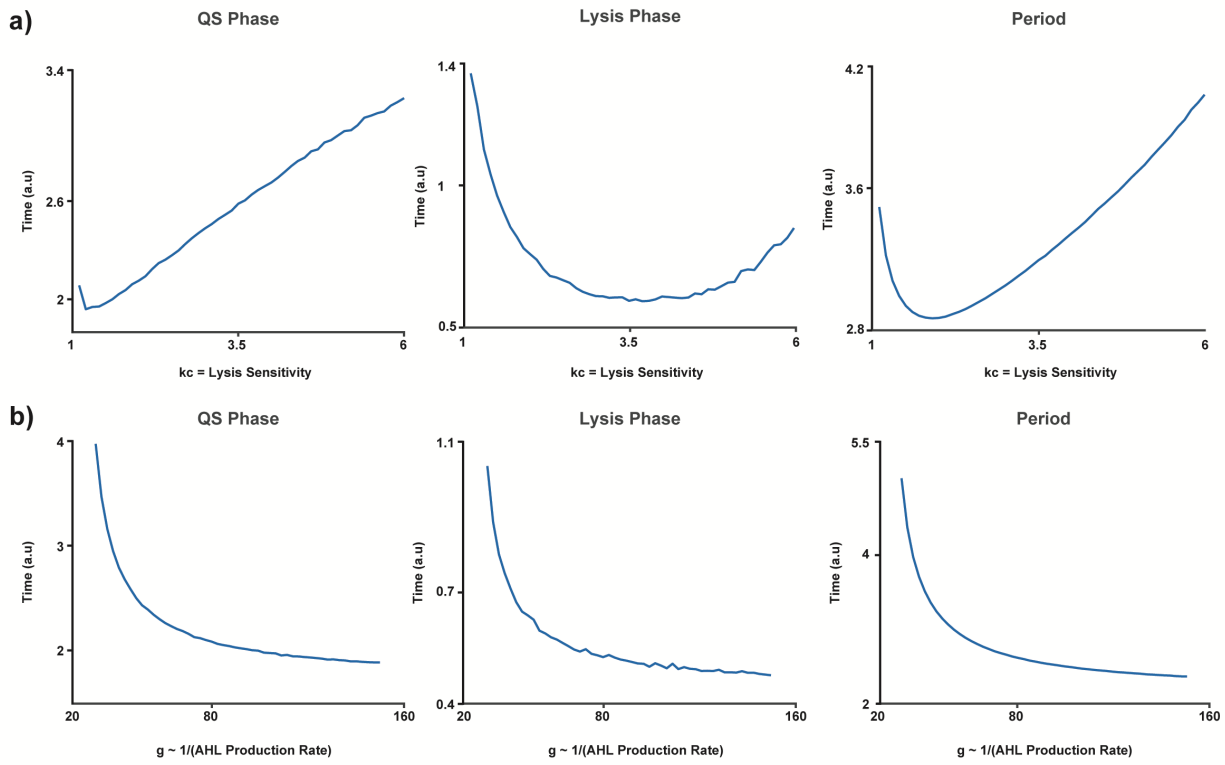


Figure S7. Simulation of lysis dynamics. **a)** QS, lysis, and cycle period changes with respect to lysis sensitivity. **b)** QS, lysis, and cycle period changes with respect to g which is proportional to inverse of AHL production.

Table S1 – All filtered mutations identified in ALE strains used in this paper. See "Table S1 – Filtered Mutations.xlsx"

Table S2 – X174E RBS sequences for each 2-plasmid SLC version used in this paper

SLC Version	X174E RBS Sequence
SLC_EcN	TAATAGATATAAATAAGGACCTTGAC
SLC_1	ATAGATATAAATAAGGACATTGTC
SLC_2	TAATAGATATAAATAAGGAATTTGAC
SLC_3	AAACGCAAGGGAGGTTGGT

Table S3 - Strains and Plasmids

Host Strain	Plasmid(s)	Figures
E. coli MG1655 WT		1
E. coli MG1655 A16_F26		1
E. coli MG1655 A16_F51		1
E. coli MG1655 A16_F89		1
E. coli Nissle WT		2, SI3
E. coli Nissle A2_F114		2, SI3
E. coli MG1655 WT	pSpSLC_RBS11	1, SI2
E. coli MG1655 A16_F26	pSpSLC_RBS11	1
E. coli MG1655 A16_F51	pSpSLC_RBS11	1
E. coli MG1655 A16_F89	pSpSLC_RBS11	1, SI2
E. coli MG1655 WT	M4	SI2
E. coli MG1655 A16_F89	M4	SI2
E. coli MG1655 WT	pTD103-sfGFP + pZA-X174E-RBS1	4
E. coli Nissle WT	pTD103-sfGFP + pZA-X174E-EcN	2, 4, SI5
E. coli Nissle A2_F114	pTD103-sfGFP + pZA-X174E-EcN	2, 3
E. coli Nissle WT	M4	SI4, SI6
E. coli Nissle A2_F114	M4	SI4, SI6
E. coli Nissle WT	pZA-X174E-RBS2	SI4, SI6
E. coli Nissle A2_F114	pZA-X174E-RBS2	SI4, SI6
E. coli Nissle WT	pTD103-sfGFP + pZA-X174E-RBS2	3, SI6
E. coli Nissle A2_F114	pTD103-sfGFP + pZA-X174E-RBS2	3, SI6
E. coli DH5alpha	pZA-X174E-EcN	3
E. coli DH5alpha	pZA-X174E-RBS1	3
E. coli DH5alpha	pZA-X174E-RBS2	3
E. coli DH5alpha	pZA-X174E-RBS3	3

Supplementary Videos

Video S1. Microfluidic dynamics of *E. coli* MG1655 wildtype, A16-F26, A16-F51, and A16-F89 strains with single plasmid SLC growing in M9 lactate medium.

Video S2. Microfluidic dynamics of *E.coli* Nissle wildtype and evolved strains with SLC-EcN growing in LB medium before and after 1mM Paraquat addition.

Video S3. Microfluidic dynamics of *E.coli* Nissle evolved strain with SLC-EcN and SLC-2, and *E.coli* Nissle wildtype strain with SLC-2, all growing in DMEM-based medium.

SLC model derivation and analysis

Deterministic model of AHL concentration can be derived as follows:

In a bacterial population within a microfluidic trap, let $a_1 = [\text{AHL}]_{\text{cell \#1}}$. The change in AHL concentration within cell #1 can be written as:

$$\frac{da_1}{dt} = \frac{Kd_T(1+\alpha c^2 a_1^2)}{1+c^2a_1^2} - k_f a_1 + k_b a_{\text{ext}} \tag{1}$$

where $\frac{Kd_T(1+\alpha c^2 a_1^2)}{1+c^2a_1^2}$ represents AHL production through positive feedback within the cell, $k_f a_1$ represents AHL diffusion out of the cell, and $k_b a_{\text{ext}}$ represents AHL diffusion into the cell.

In a population of n cells, the total external AHL can be written as:

$$\frac{da_{\text{ext}}}{dt} = k_f a_1 + k_f a_2 + ... + k_f a_n - n(k_b a_{\text{ext}}) - q a_{\text{ext}} \tag{2}$$

where qa_{ext} represents degradation of AHL due to flow of AHL out of the microfluidic trap.

Assuming all cells contain the same internal concentration of AHL:

$$\frac{da_{\text{ext}}}{dt} = n * k_f a_1 - n * k_b a_{\text{ext}} - q a_{\text{ext}} \tag{3}$$

Assuming a_{ext} reaches equilibrium quickly with respect to $\frac{da_1}{dt}$, $\frac{da_{\text{ext}}}{dt} = 0$. Thus,

$$a_{\text{ext}} = \frac{n * k_f a_1}{n * k_b + q} \tag{4}$$

$$\frac{da_1}{dt} = \frac{Kd_T(1+\alpha c^2 a_1^2)}{1+c^2a_1^2} - k_f a_1 - k_b \frac{n * k_f a_1}{n * k_b + q} \tag{5}$$

Let $a^2 = c^2 a_1^2$, $\tilde{t} = cKd_T t$, $\gamma = \frac{k_f}{cKd_T}$ (inverse of AHL production rate), $\beta = \frac{q}{k_b}$. AHL production and accumulation within the population can be simplified to:

$$\frac{da}{d\tilde{t}} = \frac{1+\alpha a^2}{1+a^2} - \frac{\gamma a \beta}{n+\beta} \tag{6}$$

Bacterial population growth is modelled as:

$$\frac{dn}{dt} = n - k_c ne \tag{7}$$

where $k_c ne$ represents cell death from lysis protein (e represents E lysis protein concentration, k_c represents lysis sensitivity).

Since AHL and E lysis protein are both driven by the *pLux* promoter, we can assume that e is approximately equal to a. Replacing $\tilde{t}$ with t in $\frac{da}{d\tilde{t}}$, the final equations for the system are as follows:

$$\frac{dn}{dt} = n - k_c na \tag{8}$$

$$\frac{da}{dt} = \frac{1 + \alpha a^2}{1 + a^2} - \frac{\gamma a \beta}{n + \beta} \tag{9}$$

The parameters used in this model are: $\alpha = 50$ (basal AHL production) and $\beta = 75$. To study effect of AHL production rate, we varied γ (inverse of AHL production) between 30 to 150 while $k_c = 3$. To study effect of lysis sensitivity, we varied k_c between 1.1 and 6 while $\gamma = 50$. To generate our final curves for MG1655, we chose $\gamma = 50$ and $k_c = 2$. These parameters yielded a QS phase of 2.11, Lysis phase of 0.76, and a period of 2.87. For Nissle, we chose $\gamma = 80$ and $k_c = 6$. These parameters yielded a QS phase of 2.63, Lysis phase of 0.76, and a period of 3.35. All model results were obtained in MATLAB using the ode23 function.

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